



HFB Joystick Series

Electronics, Firmware & Configuration Guide

Applies to	Details
Grip models	Felon, Flanker, Raptor, Hornet
Firmware	HOTAS Universal Firmware – HFB Joystick (01_HFB_Joystick.ino)
Configuration app	Tactical MFD Control Hub (Windows application, .exe)
Firmware updates	https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main
Companion document	The Assembly Instructions of your grip model (inputs layout, BOM, schematic, mechanical build)

Use this guide after the mechanical assembly is finished. It covers everything that is common to all HFB joysticks: setting the I2C addresses, verifying the wiring, flashing the firmware and configuring the joystick with the Tactical MFD Control Hub app (calibration, axis inversion, filters, special modes and haptic feedback for DCS World).

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1.- How the system works

Every HFB joystick is split into two halves connected by a 4-pin 360° pogo connector: the **base**, which holds the microcontroller and the gimbal sensors, and the **grip**, which holds the buttons, the grip sensors and the haptic motor. Both halves share one I2C bus.

Location	Component	I2C address	Function
Base	Arduino Pro Micro (ATmega32U4)	—	Main controller, USB game controller + serial port
Base	ADS1115 ADC #2	0x49	Gimbal axes X (A0) and Y (A1)
Base	3-position switch (pins 7/8), LEDs (pins 5/6)	—	Local haptic mode select, status LEDs
Grip	PCF8575 expander #1	0x20	Buttons 1–16
Grip	PCF8575 expander #2	0x21	Buttons 17–27
Grip	ADS1115 ADC #1	0x48	Twist Z (A0), mini stick Rx (A1) / Ry (A2), brake Rz (A3)
Grip	AT24C32 EEPROM	0x50	Stores the grip calibration
Grip	DRV2605L haptic driver + ERM motor	0x5A	Vibration effects

1.1.- Where your settings are stored

Data	Stored in	What it means for you
Gimbal calibration (X, Y)	Pro Micro internal EEPROM (base)	Stays with the base.
Grip calibration (Z, Rx, Ry, Rz) and disabled axes	AT24C32 EEPROM (grip)	Each grip carries its own calibration, so you can swap grips without recalibrating.
Axis inversion, filters, jitter threshold, ATB modes	Pro Micro internal EEPROM (base)	Stays with the base when you swap grips. Review it after a swap.
HFB thresholds (G / AoA)	App only (not stored)	Reset to 3.0 G / 15° every time the app starts.

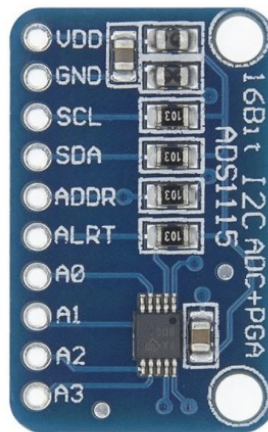
All values survive firmware updates and power cycles.

2.- I2C address configuration

Think of the Arduino as a bus driver and every chip as a passenger. All of them sit on the same two wires (**SDA** and **SCL**), so each chip needs a unique "seat number": its I2C address. This build uses two ADS1115 and two PCF8575, and identical chips leave the factory with the same address. **One chip of each pair must be re-addressed with a solder bridge before it is installed**, otherwise they conflict and neither works.

2.1.- ADS1115 analog-to-digital converters

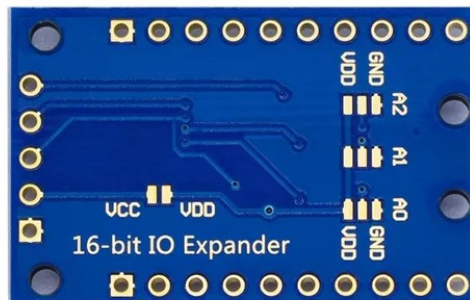
- **Grip ADS1115 – 0x48 (default):** do nothing. On these boards the ADDR pin is pulled to GND.
- **Base ADS1115 – 0x49:** bridge the **ADDR** pin to **VDD** with solder or a short wire.



ADS1115: bridge ADDR to VDD on the base unit only

2.2.- PCF8575 button expanders

- **Expander #1 – 0x20 (buttons 1–16):** leave it as it came from the factory.
- **Expander #2 – 0x21 (buttons 17–27):** on the back of the board, bridge pad **A0** to **VDD**. Some board colors/brands differ; if in doubt, check the board's datasheet and confirm with the I2C scan (section 3).



PCF8575 address pads (A0, A1, A2)

2.3.- The 4-wire backbone

All modules share the same four wires, which run from the Pro Micro through the base and the pogo connector into the grip:

Signal	Wire color	Pro Micro pin
VCC (5 V)	Red	VCC
GND	Black	GND (all GND points must be connected together)
SDA (data)	—	Pin 2
SCL (clock)	—	Pin 3

Pogo connector

The order of the four wires must match exactly on the base side and the grip side. A swapped pair is the most common cause of a grip that is not detected.

3.- Hardware verification (I2C scan)

Before flashing the joystick firmware, run a scan to confirm the wiring and the solder bridges.

1. Open `I2C_Scanner.ino` (in the FIRMWARE folder of the official repository, section 4.5) in the Arduino IDE.
2. Select **Tools** → **Board** → **Arduino Leonardo** and the joystick's port.
3. Upload the sketch with the grip **attached** to the base.
4. Open the Serial Monitor at **115200** baud.
5. The output must list: **0x20, 0x21, 0x48, 0x49, 0x50 and 0x5A**.

Result	Likely cause
Only 0x49 appears	The grip is not connected, or the pogo wires are in the wrong order.
0x48 or 0x20 missing, everything else present	Two chips share one address: the solder bridge was made on the wrong board, or not at all.
0x21 or 0x49 missing	The solder bridge is missing or did not wet both pads.
0x50 or 0x5A missing	Loose wire on that module (check VCC/GND too).
No addresses at all	SDA/SCL swapped, or a short on the bus.

4.- Firmware installation

4.1.- Required libraries

Header	Library	How to install
Joystick.h	ArduinoJoystickLibrary (Matthew Heironimus)	Download the ZIP from GitHub → Sketch → Include Library → Add .ZIP Library
PCF8575.h	PCF8575 (Rob Tillaart)	Library Manager
ADS1X15.h	ADS1X15 (Rob Tillaart)	Library Manager
Adafruit_DRV2605.h	Adafruit DRV2605 Library	Library Manager
Wire.h, EEPROM.h	Built in	—

Note

Several libraries share the name "PCF8575". Install the one by **Rob Tillaart**; the others use a different API and the sketch will not compile.

4.2.- Custom USB name and VID/PID (do this before the first upload)

By default the joystick shows up in Windows and in games as "Arduino Leonardo". Giving it its own name and USB IDs makes it easy to recognize, and keeps games from mixing up its bindings with other Arduino-based devices. Because the USB identity is compiled into the firmware, set it **before** uploading, so you only flash once.

1. Find the AVR core folder that contains `boards.txt`, usually `%LOCALAPPDATA%\Arduino15\packages\arduino\hardware\avr\1.x.x\` (the exact path may vary).
2. In that same folder, create a text file named `boards.local.txt` with the lines below. Values in this file override `boards.txt` without editing it, and you can delete the file to go back to the defaults.
3. Restart the Arduino IDE.

```
leonardo.build.vid=0x2342
leonardo.build.pid=0x8038
leonardo.build.usb_product="HFB Joystick"
```

- These settings apply to every Arduino Leonardo / Pro Micro you upload from this PC. Remove or edit the file before flashing other projects.
- Use a different PID for each device type (joystick, throttle, pedals...) so Windows treats them as separate controllers.
- After changing VID/PID, Windows detects a new device: rebind it in your games.

4.3.- Upload the firmware

1. Open `01_HFB_Joystick.ino`. Always use the latest version from the **FIRMWARE** folder of the official repository (section 4.5).
2. Select **Tools** → **Board** → **Arduino Leonardo** and the joystick's port.
3. Click **Upload**.
4. If the upload fails because the port disappears, short **RST** to **GND** on the Pro Micro and click Upload right away (some clones need two quick taps).

After the upload, the joystick shows up in Windows as a game controller (6 axes, 28 buttons, 1 hat) **and** as a serial (COM) port. The app uses that port.

4.4.- Factory defaults

On the first boot (or with a blank EEPROM) the firmware loads these defaults:

Setting	Default
Inverted axes	Y and Rz
Alpha 1 (gimbal X, Y)	0.35
Alpha 2 (grip Z, Rx, Ry, Rz)	0.15
Jitter threshold	15
ATB 1 / ATB 2	Off
Calibration	Generic values; calibrate before flying (section 5.5)

4.5.- Firmware updates

Firmware updates, fixes and new features are published in the official GitHub repository:

<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/tree/main>

1. Open the repository and download the latest firmware from the **FIRMWARE** folder (**Code** → **Download ZIP**, or open the file and use **Download raw file**).
2. Check the repository notes for changes, new required libraries or new steps before updating.
3. Close the Tactical MFD Control Hub (it uses the same COM port), then upload the new `01_HFB_Joystick.ino` exactly as in section 4.3. Leave `boards.local.txt` in place so the joystick keeps its USB name and VID/PID.
4. Reconnect with the app, press **STATUS** on the **LINK** page to see the installed version in the log (line **Firmware: ...**), and check **CTRL** > **INV** / **FILT** / **MODE** and **MON** > **INPUT**.

Your settings are kept

Calibration and settings live in EEPROM, so a normal update keeps them. Recalibrate

only if the update notes ask for it, or if the axes behave differently afterwards.

Watch the repository

On GitHub, click **Watch** → **Custom** → **Releases** to get an email when a new version is published.

5.- Tactical MFD Control Hub (configuration app)

The Tactical MFD Control Hub is a Windows desktop app styled as a fighter-jet multifunction display (MFD). From it you can:

- calibrate the gimbal and the grip,
- invert axes, tune the filters and toggle the special modes,
- drive the haptic motor in real time from DCS World telemetry (HFB),
- check every axis, hat and button.

All joystick settings are stored on the joystick itself. **You do not need the app running to fly**, except when you want HFB haptics.

5.1.- Installation and launch

The app is a single Windows program (Windows 10 or 11, 64-bit). It needs no installer and no additional software.

1. Download Tactical MFD Control Hub.exe from the **Releases** page of the official repository (<https://github.com/TheRaelNoisyBoi/DIY-HOTAS-Ecosystem/releases>) and save it in any folder, for example Documents\HOTAS. You can create a desktop shortcut to it.
2. Double-click it to start. The first start can take a few seconds while Windows unpacks the program.
3. If Windows shows "**Windows protected your PC**", click **More info** → **Run anyway**. The app is not signed with a paid code-signing certificate, so Windows SmartScreen warns about any new program. This happens only the first time.
4. If your antivirus quarantines the file, restore it and add an exception for it. Small independent .exe tools are sometimes flagged as a false positive.

Before connecting

Only one program can use the COM port at a time. **Close the Arduino IDE Serial Monitor** (and any other serial tool) before connecting from the app.

The MFD is drawn with the **Agency FB** font (installed with Microsoft Office). Without it the app still works with a default font.

5.2.- Navigating the MFD

Like a real MFD, the screen is surrounded by Option Select Buttons (OSBs). The label next to each OSB shows what it does on the current page. You can click the OSBs, or click directly on the rows, toggles and sliders shown on the screen.

OSB group	Function
Bottom row	Page selection: LINK , CTRL , MON . The active page is boxed.
Left column	Sub-pages (or the port list on LINK).

OSB group	Function
Right column	Page actions. On CTRL: UP / DN move the cursor, - / + change a slider, SEL toggles or runs the selected row.

The header strip at the top is always visible:

Flag	Lit when
GUN	DCS reports the gun firing
MSL	DCS reports a weapon release
TD	DCS reports touchdown
HFB	The real-time haptics engine is on
LINK	The app is connected to the joystick

5.3.- LINK page – connecting to the joystick



LINK page: port list (left), connection state, serial log

1. Plug in the joystick and press **SCAN** (bottom-left OSB) to refresh the port list.
2. Select the joystick's COM port with the left OSBs (or click it in the list).
3. Press **CONN**. The app silences the haptics and requests the device status.
4. When the joystick answers, **DEVICE** shows **JOYSTICK** and the app opens the CTRL page automatically.

- **STATUS** asks the joystick for its current settings again. The reply appears in the serial log.
- **DISC** closes the connection and leaves the haptics off.
- Not sure which port it is? Note the list, unplug the joystick, press SCAN, and see which port disappeared.

5.4.- CTRL page – joystick settings

Choose a sub-page with the left OSBs: **INV**, **FILT**, **MODE**, **HFB**, **CAL**. Every change is sent to the joystick and saved immediately in its memory; the app then re-reads the status to confirm. While a calibration is running, the other controls are locked.

INV – Axis inversion



Toggle **INVERT X / Y / Z / RX / RY / RZ** until every axis moves the right way in-game (check it on MON > INPUT or in Windows' game controller panel). Y and Rz are inverted by default.

FILT – Filters



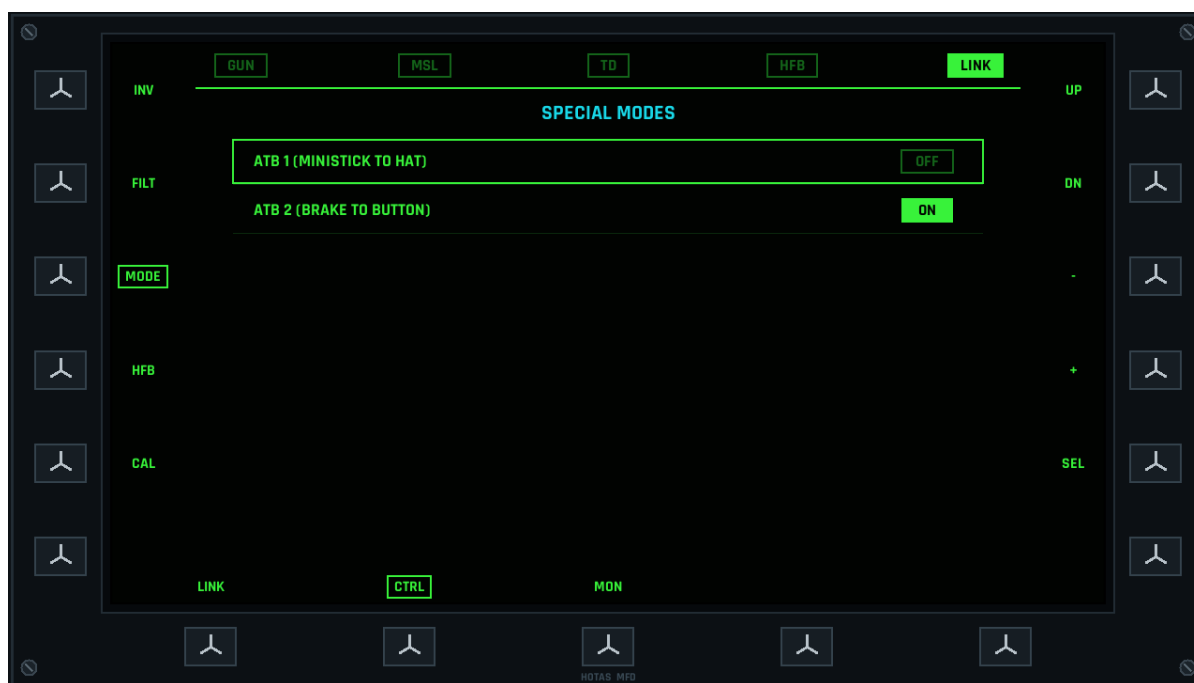
CTRL > FILT

The 16-bit sensors pick up tiny vibrations and electrical noise. The firmware smooths them with an adaptive exponential filter (EMA) and a jitter deadband:

Setting	Range	Default	Effect
ALPHA 1 (GIMBAL X,Y)	0.01 – 1.00	0.35	Lower = smoother but slower to respond. Higher = more direct, more noise.
ALPHA 2 (GRIP Z,RX,RY,RZ)	0.01 – 1.00	0.15	Same, for the grip axes.
JITTER THRESHOLD	0 – 500	15	Minimum change (output range ± 16384) before the axis value updates. Raise it if an axis flickers at rest; keep it low for precise flying.

Fast movements are not slowed down: when the stick moves quickly the filter adapts and follows it almost instantly. Drag a slider and release it, or use **- / +**; the value is sent when you release.

MODE – Special modes (Axis-To-Button)



CTRL > MODE

Mode	What it does	Models
ATB 1 (MINISTICK TO HAT)	The mini stick (Rx/Ry) stops reporting as axes and works as an 8-way hat switch.	Felon, Flanker
ATB 2 (BRAKE TO BUTTON)	The analog brake lever (Rz) stops reporting as an axis and works as button 28 (pressed beyond about 75% of travel).	Felon

The ATB LED (pin 5) lights up while either mode is on. Both modes are stored on the joystick and stay active after a power cycle.

HFB – Haptic feedback

See section 6.

5.5.- CAL – Calibration

Calibration teaches the firmware the real minimum and maximum of each sensor, so every axis uses its full range. Calibrate the gimbal once after building the base, and the grip once per grip (the grip keeps its own calibration). Recalibrate if you change a sensor, a magnet or a spring.



CTRL > CAL: choose the gimbal (base) or the grip

1. Connect on the LINK page, then go to **CTRL > CAL**.
2. Select **START CALIBRATION WIZARD**. (If HFB is on, the app switches it off first.)
3. Choose **CALIBRATE GIMBAL** (X, Y) or **CALIBRATE GRIP** (Z, RX, RY, RZ). The grip must be attached.
4. The screen names the axis being calibrated and shows its live **min / max** values. Move that axis slowly to both mechanical ends, several times, until the numbers stop changing.
5. Select **SAVE AXIS (NEXT)**. The app refuses to save a range that is too small (less than 20 counts): move the axis through its full travel and try again.
6. For an axis your model **does not have**, select **DISABLE AXIS** instead (see the table below). The axis is then held at center.
7. After the last axis the screen shows **CALIBRATION COMPLETE!** and the values are stored.
8. Repeat for the other half (gimbal or grip).



Calibrating an axis: live min/max values

Grip calibration by model

The grip wizard always walks through Z, RX, RY and RZ, in that order:

Model	Z (twist)	RX / RY (mini stick)	RZ (brake)
Felon	Calibrate	Calibrate	Calibrate
Flanker	Calibrate	Calibrate	DISABLE (the brake lever is a switch)
Raptor	Calibrate	DISABLE	DISABLE (the brake lever is a switch)
Hornet	Calibrate	DISABLE	DISABLE

Important

Disable every axis your grip doesn't have. A channel left unconnected picks up noise and shows up in games as a drifting axis.

Always calibrate **X** and **Y**. Never use DISABLE AXIS on the gimbal.

While the wizard runs, the joystick stops sending inputs to the PC. Finish it before flying.

If the app is closed in the middle of a calibration, the joystick keeps waiting. Reopen the app and reconnect (it picks up the wizard where it was), or unplug and replug the USB cable to abort.

5.6.- MON page – monitoring

DCS – Flight instruments



MON > DCS: G meter, attitude indicator, AoA, flight data and HFB intensity

This page shows the live DCS telemetry: aircraft, mission time, G-meter, attitude indicator, AoA, IAS, altitude, heading, Mach, vertical speed, the sink rate of the last touchdown, and the flight state (PARKED, TAXI, ROLL, AIRBORNE). The bar at the bottom shows the haptic intensity being sent to the joystick. The colored arcs on the G and AoA gauges start at your HFB thresholds. When there is no data, the page flashes **WAITING FOR DCS, NO DATA, DCS OFFLINE** or **EXPORT ERROR**.

INPUT – Joystick inputs



MON > INPUT: axes, hats and buttons

This is the quickest way to test the build: every axis, hat and button of the selected game controller is shown live. Use **JOY-** / **JOY+** to choose the controller and **SCAN** to refresh the list after plugging one in. The app shows buttons counted from **0**: button 0 in the app is button 1 in your model's inputs layout.

STAT – Status summary



MON > STAT

A one-page summary of the connection, the device settings, the HFB state and the DCS link, plus the raw telemetry line. Handy when you ask for support: send a screenshot of this page.

6.- Haptic feedback

The DRV2605L driver and the ERM motor in the grip support two haptic sources: **Local mode**, generated by the joystick itself and selected with the base switch, and **HFB (real-time) mode**, driven by the app from DCS telemetry.

Mode	How to enable	Haptic LED (pin 6)	Source
Off	Switch in any position other than Local, HFB off	Off	—
Local	Base switch in the position wired to pin 8	Half brightness	Stick deflection + button presses (works in any game, no PC software)
HFB (real-time)	App: CTRL > HFB > REALTIME HAPTICS ENGINE ON	Full brightness	DCS World telemetry. Takes priority over the switch.

6.1.- Local mode

- **Simulated G load:** pulling or pushing the stick past about 50% of its travel plays a vibration that gets stronger toward the end of travel (library effects 49 → 48 → 47).
- **Button feedback:** buttons 2 and 3 play a short click (effect 76); button 17 plays a strong buzz (effect 47). Check your model's inputs layout to see which controls these are.

6.2.- HFB mode with DCS World

Setting up the telemetry export

1. Close DCS World.
2. Download `HOTAS_Export.lua` from the DCS folder of the official repository (section 4.5).
3. Go to `%USERPROFILE%\Saved Games\DCS\Scripts\` (on some installs the folder is `DCS.openbeta`). Create the `Scripts` folder if it doesn't exist, and copy `HOTAS_Export.lua` into it.
4. In the same folder, open `Export.lua` with a text editor. If it doesn't exist, create an empty text file with that exact name (make sure Windows doesn't save it as `Export.lua.txt`).
5. Add the line below at the **end** of `Export.lua`, after any lines that other tools (SRS, Tacview, DCS-BIOS...) already added, and save it. The script passes the data on to those tools, so they keep working.

```
dofile(lfs.writedir() .. [[Scripts\HOTAS_Export.lua]])
```

Start DCS and enter a mission. The script writes the telemetry to `Saved Games\DCS\Temp\hotas_telemetry.txt` about 30 times per second, and the app reads it

automatically: MON > DCS should show your aircraft and live data. To uninstall, delete the dofile line and HOTAS_Export.lua.

Turning HFB on



CTRL > HFB

1. Connect to the joystick (LINK page).
2. Go to **CTRL > HFB** and set **REALTIME HAPTICS ENGINE** to **ON**. The HFB flag and the haptic LED light up.
3. Use **TEST HAPTIC PULSE** to check the motor.
4. Adjust the thresholds to taste: **G-FORCE THRESHOLD** (1.0 – 6.0 G, default 3.0) and **AOA BUFFET THRESHOLD** (5° – 35°, default 15°).

HFB reproduces these events:

Event	Haptic response
G load above the G threshold	Continuous vibration, 50 → 90%, increasing with G
AoA above the AoA threshold (buffet)	Continuous vibration, 30 → 85%, increasing with AoA
Taxiing	Continuous rumble, 50%
Takeoff / landing roll	Continuous rumble, 65%
Gun firing	85% while firing
Weapon release (missile, bomb...)	Single 0.2 s pulse, 70%
Touchdown	Single 0.3 s pulse, 60 → 95% depending on

Event	Haptic response
	sink rate

When several continuous effects apply at once, the strongest one wins.

Safety behaviour

The motor stops automatically if the telemetry stops updating (game paused, menu, crash), if no aircraft is loaded, during calibration, and when you disconnect or close the app.

The HFB thresholds are not stored. They go back to 3.0 G / 15° every time the app starts.

7.- Base controls and indicators

Item	Pin	Meaning
3-position switch	7 / 8	The position wired to pin 8 enables Local haptic mode. The other positions turn local haptics off.
ATB LED	5	On while ATB 1 or ATB 2 is active
Haptic LED	6	Off = no haptics · half brightness = Local mode · full brightness = HFB (real-time)

8.- Troubleshooting

Symptom	What to check
App: "Failed to connect"	Another program has the port open (Arduino Serial Monitor, another app instance). Close it and press CONN again.
Connected, but DEVICE stays "___"	Press STATUS. If nothing appears in the log, re-flash the firmware and check that you selected the joystick's COM port.
"Device did not answer CAL"	The joystick did not reply within 4 s. Disconnect, replug the USB cable, reconnect and try again.
Log shows "AT24C32 error." when calibrating the grip	The grip EEPROM (0x50) is not detected. Check the grip connection and run the I2C scan.
An axis drifts or moves by itself	That channel is not used by your model: disable it in the grip calibration (section 5.5).

Symptom	What to check
An axis jitters at rest	Raise the JITTER THRESHOLD, or lower the corresponding ALPHA.
Axis moves the wrong way	CTRL > INV.
Axis doesn't reach its ends, or has a dead zone at one end	Recalibrate that half (gimbal or grip) using the full mechanical travel.
MON > DCS shows WAITING FOR DCS	HOTAS_Export.lua is missing, the dofile line is not in Export.lua, or no mission is running.
MON > DCS shows EXPORT ERROR	The export script reported an error. Check Saved Games\DCS\Logs\dcsl.log.
No vibration with HFB on	Check that LINK and HFB are lit, use TEST HAPTIC PULSE, and make sure the G / AoA thresholds are not set too high.
Inputs freeze in-game	A calibration is still running on the joystick: finish it, or replug the USB cable.

Appendix A.- Serial command reference

Everything the app does can also be done from any serial terminal (for example, the Arduino Serial Monitor at **115200** baud with **Newline** line ending). Commands are not case-sensitive.

Command	Action
STATUS	Print the current settings
CAL	Open the calibration menu: 1 = base (X, Y), 2 = grip (Z, Rx, Ry, Rz). For each axis, press Enter (empty line) to save it or send d to disable it.
INV_X, INV_Y, INV_Z, INV_RX, INV_RY, INV_RZ	Toggle the inversion of that axis
ATB1, ATB2	Toggle the Axis-To-Button modes
FIL1 <0.01–1.00>	Set Alpha 1 (gimbal filter)
FIL2 <0.01–1.00>	Set Alpha 2 (grip filter)
JITTER <0–500>	Set the jitter threshold
HAPTIC_RTP_ON, HAPTIC_RTP_OFF	Enter / leave real-time haptic mode
RTP <0–127>	Set the real-time motor intensity
VCC	Diagnostic: show the measured USB supply voltage (mV)

Appendix B.- DRV2605L effect library

Reference list of the built-in DRV2605L waveforms (library 1, ERM). The numbers match the effect IDs used in section 6.1.

EFFECT ID NO.	WAVEFORM NAME	EFFECT ID NO>	WAVEFORM NAME	EFFECT ID NO.	WAVEFORM NAME
1	Strong Click - 100%	42	Long Double Sharp Click Medium 2 – 80%	83	Transition Ramp Up Long Smooth 2 – 0 to 100%
2	Strong Click - 60%	43	Long Double Sharp Click Medium 3 – 60%	84	Transition Ramp Up Medium Smooth 1 – 0 to 100%
3	Strong Click - 30%	44	Long Double Sharp Tick 1 – 100%	85	Transition Ramp Up Medium Smooth 2 – 0 to 100%
4	Sharp Click - 100%	45	Long Double Sharp Tick 2 – 80%	86	Transition Ramp Up Short Smooth 1 – 0 to 100%
5	Sharp Click - 60%	46	Long Double Sharp Tick 3 – 60%	87	Transition Ramp Up Short Smooth 2 – 0 to 100%
6	Sharp Click - 30%	47	Buzz 1 – 100%	88	Transition Ramp Up Long Sharp 1 – 0 to 100%
7	Soft Bump - 100%	48	Buzz 2 – 80%	89	Transition Ramp Up Long Sharp 2 – 0 to 100%
8	Soft Bump - 60%	49	Buzz 3 – 60%	90	Transition Ramp Up Medium Sharp 1 – 0 to 100%
9	Soft Bump - 30%	50	Buzz 4 – 40%	91	Transition Ramp Up Medium Sharp 2 – 0 to 100%
10	Double Click - 100%	51	Buzz 5 – 20%	92	Transition Ramp Up Short Sharp 1 – 0 to 100%
11	Double Click - 60%	52	Pulsing Strong 1 – 100%	93	Transition Ramp Up Short Sharp 2 – 0 to 100%
12	Triple Click - 100%	53	Pulsing Strong 2 – 60%	94	Transition Ramp Down Long Smooth 1 – 50 to 0%
13	Soft Fuzz - 60%	54	Pulsing Medium 1 – 100%	95	Transition Ramp Down Long Smooth 2 – 50 to 0%
14	Strong Buzz - 100%	55	Pulsing Medium 2 – 60%	96	Transition Ramp Down Medium Smooth 1 – 50 to 0%
15	750 ms Alert 100%	56	Pulsing Sharp 1 – 100%	97	Transition Ramp Down Medium Smooth 2 – 50 to 0%
16	1000 ms Alert 100%	57	Pulsing Sharp 2 – 60%	98	Transition Ramp Down Short Smooth 1 – 50 to 0%
17	Strong Click 1 - 100%	58	Transition Click 1 – 100%	99	Transition Ramp Down Short Smooth 2 – 50 to 0%
18	Strong Click 2 - 80%	59	Transition Click 2 – 80%	100	Transition Ramp Down Long Sharp 1 – 50 to 0%
19	Strong Click 3 - 60%	60	Transition Click 3 – 60%	101	Transition Ramp Down Long Sharp 2 – 50 to 0%
20	Strong Click 4 - 30%	61	Transition Click 4 – 40%	102	Transition Ramp Down Medium Sharp 1 – 50 to 0%
21	Medium Click 1 - 100%	62	Transition Click 5 – 20%	103	Transition Ramp Down Medium Sharp 2 – 50 to 0%
22	Medium Click 2 - 80%	63	Transition Click 6 – 10%	104	Transition Ramp Down Short Sharp 1 – 50 to 0%
23	Medium Click 3 - 60%	64	Transition Hum 1 – 100%	105	Transition Ramp Down Short Sharp 2 – 50 to 0%
24	Sharp Tick 1 - 100%	65	Transition Hum 2 – 80%	106	Transition Ramp Up Long Smooth 1 – 0 to 50%
25	Sharp Tick 2 - 80%	66	Transition Hum 3 – 60%	107	Transition Ramp Up Long Smooth 2 – 0 to 50%
26	Sharp Tick 3 – 60%	67	Transition Hum 4 – 40%	108	Transition Ramp Up Medium Smooth 1 – 0 to 50%
27	Short Double Click Strong 1 – 100%	68	Transition Hum 5 – 20%	109	Transition Ramp Up Medium Smooth 2 – 0 to 50%
28	Short Double Click Strong 2 – 80%	69	Transition Hum 6 – 10%	110	Transition Ramp Up Short Smooth 1 – 0 to 50%
29	Short Double Click Strong 3 – 60%	70	Transition Ramp Down Long Smooth 1 – 100 to 0%	111	Transition Ramp Up Short Smooth 2 – 0 to 50%
30	Short Double Click Strong 4 – 30%	71	Transition Ramp Down Long Smooth 2 – 100 to 0%	112	Transition Ramp Up Long Sharp 1 – 0 to 50%
31	Short Double Click Medium 1 – 100%	72	Transition Ramp Down Medium Smooth 1 – 100 to 0%	113	Transition Ramp Up Long Sharp 2 – 0 to 50%
32	Short Double Click Medium 2 – 80%	73	Transition Ramp Down Medium Smooth 2 – 100 to 0%	114	Transition Ramp Up Medium Sharp 1 – 0 to 50%
33	Short Double Click Medium 3 – 60%	74	Transition Ramp Down Short Smooth 1 – 100 to 0%	115	Transition Ramp Up Medium Sharp 2 – 0 to 50%

EFFECT ID NO.	WAVEFORM NAME	EFFECT ID NO>	WAVEFORM NAME	EFFECT ID NO.	WAVEFORM NAME
34	Short Double Sharp Tick 1 – 100%	75	Transition Ramp Down Short Smooth 2 – 100 to 0%	116	Transition Ramp Up Short Sharp 1 – 0 to 50%
35	Short Double Sharp Tick 2 – 80%	76	Transition Ramp Down Long Sharp 1 – 100 to 0%	117	Transition Ramp Up Short Sharp 2 – 0 to 50%
36	Short Double Sharp Tick 3 – 60%	77	Transition Ramp Down Long Sharp 2 – 100 to 0%	118	Long buzz for programmatic stopping – 100%
37	Long Double Sharp Click Strong 1 – 100%	78	Transition Ramp Down Medium Sharp 1 – 100 to 0%	119	Smooth Hum 1 (No kick or brake pulse) – 50%
38	Long Double Sharp Click Strong 2 – 80%	79	Transition Ramp Down Medium Sharp 2 – 100 to 0%	120	Smooth Hum 2 (No kick or brake pulse) – 40%
39	Long Double Sharp Click Strong 3 – 60%	80	Transition Ramp Down Short Sharp 1 – 100 to 0%	121	Smooth Hum 3 (No kick or brake pulse) – 30%
40	Long Double Sharp Click Strong 4 – 30%	81	Transition Ramp Down Short Sharp 2 – 100 to 0%	122	Smooth Hum 4 (No kick or brake pulse) – 20%
41	Long Double Sharp Click Medium 1 – 100%	82	Transition Ramp Up Long Smooth 1 – 0 to 100%	123	Smooth Hum 5 (No kick or brake pulse) – 10%

For questions contact me at: the.noisy.boi@gmail.com

ALL DONE – HAPPY FLYING!